

Utsab Sinha

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Kolkata, West Bengal, India

Summary

Computer Science undergraduate (CGPA 8.29/10) with hands-on full-stack and software development experience, including building and shipping production-grade features with React.js, Node.js, and REST APIs across live platforms. Complements this with applied machine learning and NLP experience, including a full-stack AI Brand Intelligence System processing 10,000+ social posts per query run using VADER sentiment classification and LDA topic modelling. Seeking software development or data science internship/full-time roles.

Education

University of Engineering and Management (UEM), Jaipur <i>B.Tech in Computer Science Engineering – SGPA: 8.17/10.0</i>	Jaipur, Rajasthan <i>July 2023 – May 2027</i>
New Andul H.C. School <i>WBCHSE (Class XII) – 82.40%</i>	Howrah, West Bengal <i>2021</i>
New Andul H.C. School <i>WBBSE (Class X) – 77.57%</i>	Howrah, West Bengal <i>2019</i>

Technical Skills

Web & Databases: React.js, Node.js, REST APIs, MySQL, PostgreSQL, MongoDB
Core CS & Systems: Object-Oriented Programming (Java), Data Structures (Trees, Graphs), Operating Systems, Computer Networking, Software Engineering Fundamentals, Cloud Computing, Cybersecurity Fundamentals
Languages & Tools: Python, Java, C++, SQL, Git, Docker, Tweepy, PRAW
Machine Learning: Predictive Modelling, scikit-learn, PyTorch, Computer Vision (OpenCV), SMOTE, GridSearchCV, Random Forest, Gradient Boosting
NLP & AI: Sentiment Analysis (VADER, TextBlob), LDA Topic Modelling, Generative AI & LLM Fundamentals, Social Media Mining (Twitter/Reddit APIs)
Data & BI Tools: Tableau, Streamlit, pandas, NumPy, Jupyter Notebook, EDA

Professional Experience

AI/ML Intern <i>FlyRank AI (Remote)</i>	July 2026 – Present
- Working on the ML layer of the product – embeddings, clustering, and intent classification – to power search and recommendation features beyond API-level integration. - Collaborating with the product team to translate model outputs into user-facing insights and iterate on model performance.	
AI/ML & Business Intelligence Intern <i>Twidix (Remote)</i>	April 2026 – July 2026
- Built and shipped full-stack AI/ML features for a live production platform, including 8+ consumer-analytics dashboards, predictive insight tools, and workflow automation modules used by 150+ end users. - Implemented data pipelines integrated into production using REST APIs and Git, partnering directly with the product team to lift key engagement metrics by 15%.	
Data Analyst Intern <i>Bluestock Fintech (Remote)</i>	April 2026 – May 2026
Analyzed 5+ financial and market datasets to support data-driven decision-making, cutting manual data-cleaning time by 25% through structured exploration and reporting workflows.	

Projects

Antar AI – Real-Time Engagement & Focus Platform — <i>React, TypeScript, Computer Vision</i>	2025
- Engineered a fully client-side computer vision pipeline (no external ML model or API) that detects faces and tracks pupils using custom RGB skin-tone and luminance-minimization heuristics on a downscaled 120x90 canvas, achieving sub-1ms frame processing at under 0.1% CPU usage. - Built a real-time engagement and cognitive focus scoring engine combining gaze deviation, head-pose stability, blink-rate (EAR) fatigue detection, and a multi-factor confidence model, with adaptive scoring weights across 4 activity modes (study, meeting, coding, gaming).	

- Implemented signal-processing techniques including Kalman filter jitter stabilization, exponential (IIR) coordinate smoothing, Bayesian focus-probability estimation, and webcam-based photoplethysmography (PPG) for real-time heart rate and HRV extraction.
- Solved browser tab-throttling by offloading session polling to a dedicated Web Worker ticking every 300ms, with crash-safe session auto-recovery via local persistence for uninterrupted long-running tracking.

AI Brand Intelligence System — *Python, NLP, LDA, Streamlit, React.js, Tweepy, PRAW* 2026

- Built a full-stack brand monitoring platform ingesting live Twitter and Reddit data, processing 10,000+ posts per query run through a 3-stage NLP pipeline (VADER/TextBlob sentiment, LDA topic modelling across 10 themes, keyword trend tracking).
- Architected the end-to-end full-stack build – a Node.js/REST API backend feeding a 4-module React.js and Streamlit dashboard – automating detection of top 5 consumer complaints and brand risks per cycle and cutting manual monitoring effort by 70%.

Consumer Booking Behavior Prediction — *Python, scikit-learn, pandas, NumPy* 2025

- Built an end-to-end ML pipeline on a 50,000-record airline dataset, engineering 15+ features and benchmarking 3 classifiers to achieve a cross-validated ROC-AUC of 0.88 (12% above baseline).
- Corrected a 4:1 class imbalance with SMOTE and tuned 8 hyperparameters via GridSearchCV, producing a feature-importance report ranking the top 7 purchase-intent drivers.

GenAuth AI – Real-Time Face Recognition System — *Python, OpenCV, PyTorch, CNN* 2026

- Developing a CNN-based biometric authentication pipeline achieving sub-100ms inference latency across 500+ test frames from live webcam feeds.
- Applying 4 data augmentation techniques (rotation, brightness shift, occlusion simulation, horizontal flip) to improve out-of-distribution accuracy by 11%.

Leadership

Event Coordinator, AceHack 4.0 National Hackathon 2024 – 2025

UEM Jaipur

- Led operations for a 500+ participant hackathon across 12 colleges with a 20-person volunteer team, redesigning logistics workflows to cut wait times by 25%.

Marketing Executive & Software Developer, Pradyog Club 2024 – 2025

UEM Jaipur

- Organized and promoted 15+ club events and campaigns as Marketing Executive, driving turnout through on-campus outreach and social media promotion.
- Built and maintained an event registration and management tool for the club, handling 200+ student registrations per event and streamlining check-in for organizers. Also served as a connecting member representing college.

Certifications & Coursework

Software Development & Programming

- Introduction to Software Engineering – IBM | Introduction to Object-Oriented Programming with Java – LearnQuest | Introduction to Java – LearnQuest | Trees and Graphs: Basics – University of Colorado Boulder

Systems, Cloud & Security

- Cloud Computing Fundamentals – University of Pittsburgh | Intro to Operating Systems 2: Memory Management – Codio | The Bits and Bytes of Computer Networking – Google | Cyber Security Fundamentals – University of London | Databases and SQL for Data Science with Python – IBM

AI, ML & Data

- Generative AI Fundamentals Specialization – IBM | Natural Language Processing with Probabilistic Models – DeepLearning.AI | Introduction to Data Analytics – IBM

Professional Practice

- Agile Project Management – University of Colorado Boulder

Additional: App Development Certification – EduSkills Foundation (2023) | Data Analytics Virtual Internship – Deloitte (2024) | Data Science Virtual Internship – British Airways (2024) | Quantitative Research Virtual Internship – J.P. Morgan (2024)